

A Reflected Two-Channel Quotient Obstruction

A finite derivative-certificate reduction for inner-strip zeta zeros

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Abstract

This paper develops the inner-strip continuation of the Mellin-symmetric quotient-obstruction method. The outer-strip argument proves a high-height zero-free region by comparing a quotient numerator

$$\mathcal{N}_Q(s)$$

with an effective bracket

$$\frac{i}{\tau} \mathfrak{B}_{\text{eff}}(\sigma, \tau).$$

Away from the critical line, the bracket has a fixed positive lower bound. Near the critical line, however, this one-channel lower bound degenerates. Thus the outer-strip proof cannot be pushed to $\Re s = 1/2$ by merely tightening constants.

The inner-strip mechanism is different. If

$$s_- = \frac{1}{2} - u + i\tau$$

is an off-critical zero, then the functional equation and conjugation symmetry force the reflected point

$$s_+ = \frac{1}{2} + u + i\tau$$

to be a zero as well. Hence an off-critical reflected zero pair forces two quotient equations:

$$\mathcal{N}_Q(s_-) = 0, \quad \mathcal{N}_Q(s_+) = 0.$$

Equivalently, it forces both the reflected average and the reflected finite difference to vanish. This leads to the two-channel detector

$$\mathfrak{D}(u, \tau) = \left(\frac{\mathcal{N}_Q(s_-) + \mathcal{N}_Q(s_+)}{2}, \frac{\mathcal{N}_Q(s_-) - \mathcal{N}_Q(s_+)}{2u} \right).$$

The main structural observation is that the corresponding bracket vector does not degenerate at the critical line. Its limiting components are

$$\mathfrak{B}_{\text{eff}}\left(\frac{1}{2}, \tau\right) \quad \text{and} \quad -\partial_\sigma \mathfrak{B}_{\text{eff}}\left(\frac{1}{2}, \tau\right),$$

and these two quantities cannot vanish simultaneously. In fact, the critical line bracket vector has the explicit lower scale

$$\frac{\sqrt{2}}{48}(23 \log 2 - 2) > 0.$$

For a finite inner strip

$$0 < \left| \Re s - \frac{1}{2} \right| \leq 0.01,$$

this lower bound is replaced by a finite interval certificate, for which the conservative public target is

$$g_0 = 0.2.$$

The residual side requires one new ingredient beyond the outer-strip certificate: each residual sector must be certified not only in value but also in its σ -derivative. The ordinary residual controls the reflected average, while the derivative residual controls the divided reflected difference by the mean value theorem. Thus the inner-strip problem is reduced to a finite certificate for the seven residual sectors and their σ -derivatives.

This paper is a structural reduction. It does not by itself claim a completed inner-strip zero-free theorem. The certified theorem is obtained only after the public file

INNER_STRIP_CERTIFICATE_SAFE.json

has status

inner_full_certificate.

Contents

1	Introduction	3
2	Structural input from the outer-strip quotient framework	6
2.1	Imported quotient expansion	7
2.2	Effective bracket	7
2.3	Residual norms required for the inner strip	8
2.4	Certification interface	9
3	Reflected zero pairs	10
4	The two-channel detector	11
4.1	Bracket and residual channels	12
4.2	The contradiction mechanism	14
5	Bracket-vector lower bound	14
5.1	Critical-line bracket value and derivative	15
5.2	Critical-line vector nondegeneracy	16
5.3	Finite-width bracket certificate	18
6	Residual-vector bound	19
6.1	Sector-level assumptions	19
6.2	Even residual channel	20
6.3	Divided odd residual channel	21
6.4	Residual-vector bound	22
6.5	Certificate form of the residual-vector theorem	23
7	Finite certifiability of the derivative residual sectors	23
7.1	General derivative principle	24
7.2	Algebraic coefficient derivatives	24
7.3	Compact-boundary sector	25
7.4	Compact-complement sectors	26
7.5	Band sector	27
7.6	Endpoint sector	27
7.7	Far stationary sector	28
7.8	Always-nonstationary sector	29

7.9 Floor and core sectors	29
7.10 Summary of derivative certifiability	30
8 Certified inner-strip exclusion	30
9 Height bridge and public numerical target	32
10 Public certificate interface	34
10.1 Reproduction commands	35
11 Combined consequence with the certified outer strip	35

1 Introduction

The goal of this paper is to isolate the correct inner-strip obstruction for possible off-critical zeros of the Riemann zeta function. It is a continuation of the outer-strip quotient-obstruction framework developed in the structural outer-strip paper and certified in the companion finite certificate. The outer-strip mechanism is based on an asymptotic identity of the form

$$\mathcal{N}_Q(s) = \frac{i}{\tau} \mathfrak{B}_{\text{eff}}(\sigma, \tau) + \mathcal{R}_{\text{total}}(s), \quad s = \sigma + i\tau,$$

where $\mathcal{N}_Q(s)$ is a quotient-normalized numerator, $\mathfrak{B}_{\text{eff}}(\sigma, \tau)$ is an effective bracket, and $\mathcal{R}_{\text{total}}(s)$ is the grouped residual.

In a fixed outer strip, for example

$$0.01 \leq \sigma \leq 0.49,$$

the bracket satisfies a positive lower bound

$$|\mathfrak{B}_{\text{eff}}(\sigma, \tau)| \geq d > 0.$$

The residual is certified to satisfy

$$|\mathcal{R}_{\text{total}}(s)| \leq C \frac{\log \tau}{\tau^{3/2}}.$$

Thus the main term has size at least d/τ , while the residual has size $O(\log \tau / \tau^{3/2})$. For sufficiently large τ , the main term dominates. Since every zero satisfies

$$\mathcal{N}_Q(s) = 0,$$

this gives a contradiction and therefore excludes zeros in the fixed outer strip.

This argument is deliberately an outer-strip argument. It cannot be extended to the critical line by merely improving the constant C . The reason is structural: the one-point bracket lower bound degenerates as

$$\sigma \rightarrow \frac{1}{2}.$$

If

$$u = \frac{1}{2} - \sigma,$$

then the one-channel bracket gap is of first order in u . Consequently the outer-strip comparison can at best see a main term of size comparable to

$$\frac{u}{\tau}.$$

A residual estimate of size

$$\frac{\log \tau}{\tau^{3/2}}$$

does not by itself exclude arbitrarily small positive u . Therefore the inner strip requires a different detector.

The correct detector comes from the symmetry of the zeros. Suppose

$$s_- = \frac{1}{2} - u + i\tau, \quad 0 < u \leq u_0,$$

is a nontrivial zero. By the functional equation and conjugation symmetry, the reflected point

$$s_+ = \frac{1}{2} + u + i\tau$$

is also a nontrivial zero. Hence the zero condition gives two exact equations:

$$\mathcal{N}_Q(s_-) = 0, \quad \mathcal{N}_Q(s_+) = 0.$$

It is therefore wasteful to test only one value of $\mathcal{N}_Q$. The reflected pair forces both the average

$$\frac{\mathcal{N}_Q(s_-) + \mathcal{N}_Q(s_+)}{2}$$

and the divided difference

$$\frac{\mathcal{N}_Q(s_-) - \mathcal{N}_Q(s_+)}{2u}$$

to vanish.

This is the key inner-strip idea. The outer-strip bracket weakens at the critical line, but an off-critical reflected pair is more rigid than a single point: it must make both the reflected midpoint value and the reflected slope vanish. The two-channel detector records precisely these two quantities.

We therefore define

$$\mathfrak{D}(u, \tau) := (\mathcal{E}(u, \tau), \mathcal{O}(u, \tau)),$$

where

$$\mathcal{E}(u, \tau) := \frac{\mathcal{N}_Q(s_-) + \mathcal{N}_Q(s_+)}{2},$$

and

$$\mathcal{O}(u, \tau) := \frac{\mathcal{N}_Q(s_-) - \mathcal{N}_Q(s_+)}{2u}.$$

At an off-critical reflected zero pair,

$$\mathfrak{D}(u, \tau) = (0, 0).$$

Substituting the structural quotient expansion into the two channels gives

$$\mathfrak{D}(u, \tau) = \frac{i}{\tau} \mathfrak{B}(u, \tau) + \mathfrak{R}(u, \tau),$$

where $\mathfrak{B}$ is the reflected bracket vector and $\mathfrak{R}$ is the reflected residual vector. The bracket vector is

$$\mathfrak{B}(u, \tau) = (B_{\text{ev}}(u, \tau), B_{\text{od}}(u, \tau)),$$

with

$$B_{\text{ev}}(u, \tau) = \frac{\mathfrak{B}_{\text{eff}}(\frac{1}{2} - u, \tau) + \mathfrak{B}_{\text{eff}}(\frac{1}{2} + u, \tau)}{2},$$

and

$$B_{\text{od}}(u, \tau) = \frac{\mathfrak{B}_{\text{eff}}(\frac{1}{2} - u, \tau) - \mathfrak{B}_{\text{eff}}(\frac{1}{2} + u, \tau)}{2u}.$$

As $u \rightarrow 0$, these components tend to

$$\mathfrak{B}_{\text{eff}}\left(\frac{1}{2}, \tau\right) \quad \text{and} \quad -\partial_{\sigma}\mathfrak{B}_{\text{eff}}\left(\frac{1}{2}, \tau\right).$$

The crucial point is that these limiting components cannot vanish simultaneously. The reflected even channel sees the value of the bracket on the critical line; the reflected odd channel sees its transverse derivative.

This is the inner-strip replacement for the outer-strip bracket gap. Instead of proving

$$|\mathfrak{B}_{\text{eff}}(\sigma, \tau)| \geq d$$

at one point, we prove a vector lower bound

$$\|\mathfrak{B}(u, \tau)\| \geq g_0.$$

For the final certified inner strip

$$0 < u \leq 0.01,$$

the conservative target is

$$g_0 = 0.2.$$

The residual side also changes. The even residual is controlled by the ordinary residual bound. The divided odd residual is controlled by a σ -derivative bound. Thus each residual sector must be certified in two norms:

$$|\mathcal{R}_{\nu}(s)| \leq C_{\nu,0} \frac{\log \tau}{\tau^{3/2}},$$

and

$$|\partial_{\sigma}\mathcal{R}_{\nu}(s)| \leq C_{\nu,1} \frac{\log \tau}{\tau^{3/2}}.$$

The seven sectors are the same sectors as in the outer-strip decomposition:

$$\text{band, } \text{bd, } \text{far, } \text{end, } \text{nonstat, } \text{floor, } \text{core.}$$

The derivative certificate is what makes the divided reflected difference finite and uniform as $u \rightarrow 0$.

The resulting exclusion mechanism is simple. If

$$\|\mathfrak{B}(u, \tau)\| \geq g_0$$

and

$$\|\mathfrak{R}(u, \tau)\| \leq C_{\text{inner}} \frac{\log \tau}{\tau^{3/2}},$$

then

$$\|\mathfrak{D}(u, \tau)\| \geq \frac{g_0}{\tau} - C_{\text{inner}} \frac{\log \tau}{\tau^{3/2}}.$$

This is positive whenever

$$\sqrt{\tau} > \frac{C_{\text{inner}}}{g_0} \log \tau.$$

But a reflected off-critical zero pair would force

$$\mathfrak{D}(u, \tau) = (0, 0).$$

Hence no such pair can exist above the certified height.

The paper is organized as follows. In [Section 2](#), we state the structural quotient expansion imported from the outer-strip paper. In [Section 3](#), we record the exact reflected zero-pair identity. In [Section 4](#), we define the two-channel detector and derive its bracket-residual decomposition. In [Section 5](#), we prove the algebraic nondegeneracy of the critical-line bracket vector and state the finite inner-strip bracket certificate. In [Section 6](#), we show how ordinary and σ -derivative residual certificates imply the two-channel residual bound. In [Section 7](#), we explain sector by sector why the derivative constants are finitely certifiable. Finally, in [Section 8](#), we state the certified inner-strip exclusion theorem conditional on the final public certificate

INNER_STRIP_CERTIFICATE_SAFE.json.

Remark 1.1 (Scope). *This paper is an inner-strip structural reduction. It does not claim a completed inner-strip zero-free theorem until the derivative-aware certificate has been produced and the final JSON file has status*

inner_full_certificate.

The purpose of the paper is to prove that the inner-strip problem reduces to a finite list of ordinary and σ -derivative residual certificates.

2 Structural input from the outer-strip quotient framework

This section records the analytic input imported from the outer-strip quotient-obstruction framework. The inner-strip argument does not rederive the Mellin identities, the four-family decomposition, the Poisson band expansion, or the compact-boundary construction. Those belong to the structural outer-strip paper. What changes here is the way in which the already-derived quotient expansion is tested near the critical line.

Throughout this paper we write

$$s = \sigma + i\tau, \quad \tau > 0,$$

and we use the quotient numerator

$$\mathcal{N}_Q(s)$$

from the outer-strip paper. At every nontrivial zero of $\zeta(s)$,

$$\mathcal{N}_Q(s) = 0.$$

The precise normalization used here is the same as in the outer-strip paper:

$$\mathcal{N}_Q(s) = B(s) + \frac{s-1}{s}A(s),$$

where

$$A(s) := \int_1^\infty \{x\}x^{-s-1} dx,$$

and

$$B(s) := \int_1^\infty \{x\}x^{s-2} dx.$$

Equivalently, one may use the multiplied numerator

$$sB(s) + (s-1)A(s),$$

since $s \neq 0$ in the critical strip. In this paper we use the quotient normalization $\mathcal{N}_Q$, because it is the normalization in which the effective bracket appears with prefactor i/τ .

2.1 Imported quotient expansion

The structural outer-strip paper proves an expansion of the form

$$\mathcal{N}_Q(s) = \frac{i}{\tau} \mathfrak{B}_{\text{eff}}(\sigma, \tau) + \mathcal{R}_{\text{total}}(s),$$

where $\mathfrak{B}_{\text{eff}}(\sigma, \tau)$ is the effective quotient bracket and $\mathcal{R}_{\text{total}}(s)$ is the total grouped residual.

For the inner-strip argument we require this expansion on the symmetric neighborhood

$$\left| \sigma - \frac{1}{2} \right| \leq 0.01, \quad \tau \geq T_0,$$

where in the certified implementation one may take

$$T_0 = 10.$$

The logarithm in all residual norms is the natural logarithm.

Assumption 2.1 (Imported quotient expansion on the inner strip). *On the region*

$$\left| \sigma - \frac{1}{2} \right| \leq 0.01, \quad \tau \geq T_0,$$

the quotient numerator admits the decomposition

$$\mathcal{N}_Q(s) = \frac{i}{\tau} \mathfrak{B}_{\text{eff}}(\sigma, \tau) + \mathcal{R}_{\text{total}}(s).$$

The total residual decomposes into the seven sectors

$$\mathcal{R}_{\text{total}}(s) = \mathcal{R}_{\text{band}}(s) + \mathcal{R}_{\text{bd}}(s) + \mathcal{R}_{\text{far}}(s) + \mathcal{R}_{\text{end}}(s) + \mathcal{R}_{\text{nonstat}}(s) + \mathcal{R}_{\text{floor}}(s) + \mathcal{R}_{\text{core}}(s).$$

The sector names and normalizations are the same as in the outer-strip structural paper and its certification companion.

Remark 2.2 (Why this is an assumption here). *The present paper is the inner-strip structural reduction. It is not intended to repeat the analytic derivation of the quotient expansion. The role of [Assumption 2.1](#) is to state the exact interface required from the outer-strip structural paper. The new work here begins after this interface: we replace the one-channel outer-strip test by a reflected two-channel test and identify the additional derivative residual certificates needed near the critical line.*

2.2 Effective bracket

The effective bracket has the form

$$\begin{aligned} \mathfrak{B}_{\text{eff}}(\sigma, \tau) = & 2^\sigma e^{iL} \left(\frac{12 - \sigma}{48} + \frac{iX}{4} \right) \\ & - 2^{-\sigma} e^{-iL} \left(\frac{11 + \sigma}{24} - \frac{iX}{2} \right), \end{aligned}$$

where

$$L := \tau \log 2,$$

and

$$X := \Xi_{\text{end}}(\tau)$$

is the real regularized endpoint scalar imported from the structural four-family endpoint analysis.

The only properties of X used in the bracket-vector argument are that

$$X \in \mathbb{R}$$

and that the same value of X appears in the reflected evaluations

$$\mathfrak{B}_{\text{eff}}\left(\frac{1}{2} - u, \tau\right), \quad \mathfrak{B}_{\text{eff}}\left(\frac{1}{2} + u, \tau\right).$$

The critical bracket-vector lower bound proved later is uniform in real X ; therefore no numerical upper bound for X is needed in the algebraic critical-line nondegeneracy argument.

Remark 2.3 (Why no bound for X is needed at the critical line). *The outer-strip proof uses a one-channel modulus lower bound for $\mathfrak{B}_{\text{eff}}(\sigma, \tau)$. That lower bound depends on comparing two amplitudes and degenerates as $\sigma \rightarrow 1/2$. The inner-strip proof instead uses the two quantities*

$$\mathfrak{B}_{\text{eff}}\left(\frac{1}{2}, \tau\right) \quad \text{and} \quad \partial_{\sigma} \mathfrak{B}_{\text{eff}}\left(\frac{1}{2}, \tau\right).$$

Their combined squared norm has a positive lower bound uniformly for every real value of X . This is the algebraic reason the reflected two-channel detector is the correct inner-strip object.

2.3 Residual norms required for the inner strip

The outer-strip certificate controls the residual itself in the norm

$$\frac{\log \tau}{\tau^{3/2}}.$$

For the inner strip, this is not enough. The reflected average channel uses the ordinary residual, but the divided odd channel contains

$$\frac{\mathcal{R}_{\text{total}}(s_-) - \mathcal{R}_{\text{total}}(s_+)}{2u}.$$

To control this uniformly as $u \rightarrow 0$, one needs a uniform σ -derivative bound for the residual.

Therefore each residual sector must be certified in two norms.

Definition 2.4 (Inner-strip value and derivative sector constants). *Let*

$$\nu \in \{\text{band}, \text{bd}, \text{far}, \text{end}, \text{nonstat}, \text{floor}, \text{core}\}.$$

A pair of constants

$$C_{\nu,0}, \quad C_{\nu,1}$$

certifies the sector $\mathcal{R}_{\nu}$ on the inner strip if, for every

$$\left| \sigma - \frac{1}{2} \right| \leq 0.01, \quad \tau \geq T_0,$$

one has

$$|\mathcal{R}_{\nu}(\sigma + i\tau)| \leq C_{\nu,0} \frac{\log \tau}{\tau^{3/2}},$$

and

$$|\partial_{\sigma} \mathcal{R}_{\nu}(\sigma + i\tau)| \leq C_{\nu,1} \frac{\log \tau}{\tau^{3/2}}.$$

Define the total value and derivative constants by

$$C_0 := \sum_{\nu} C_{\nu,0},$$

and

$$C_1 := \sum_{\nu} C_{\nu,1}.$$

Finally define the inner vector residual constant

$$C_{\text{inner}} := \sqrt{C_0^2 + C_1^2}.$$

Remark 2.5 (Why the derivative norm is the right one). *The divided odd channel contains a difference quotient in the real direction. For*

$$s_{\pm} = \frac{1}{2} \pm u + i\tau,$$

the mean value theorem gives

$$\left| \frac{\mathcal{R}_{\text{total}}(s_-) - \mathcal{R}_{\text{total}}(s_+)}{2u} \right| \leq \sup_{|\sigma - \frac{1}{2}| \leq u} |\partial_{\sigma} \mathcal{R}_{\text{total}}(\sigma + i\tau)|.$$

Thus a uniform ∂_{σ} -residual certificate is exactly what is needed to make the divided odd channel finite and stable as $u \rightarrow 0$.

2.4 Certification interface

The final inner-strip theorem will not cite individual hand-transcribed constants. Instead, it will cite a public JSON certificate. The required certificate contract is as follows.

For each sector ν , the corresponding JSON file must have status

`proved`

and must contain the two fields

`C_value_upper_safe,` `C_sigma_derivative_upper_safe.`

The final merge file

`INNER_STRIP_CERTIFICATE_SAFE.json`

must have status

`inner_full_certificate.`

It must record

$g_0,$ $C_0,$ $C_1,$ $C_{\text{inner}},$

and the verified height inequality

$$\sqrt{T} > \frac{C_{\text{inner}}}{g_0} \log T.$$

For the public target used in this paper, the bracket-vector certificate will use

$$g_0 = 0.2.$$

If the final merged certificate verifies

$$T = 10^{30},$$

then the inner-strip exclusion height matches the outer-strip certified height.

Remark 2.6 (No hidden constants). *All constants used in the final inner-strip theorem must appear either in the structural formulas of this paper or in the final public certificate file. There is no untracked residual sector and no untracked derivative sector.*

3 Reflected zero pairs

The inner-strip argument begins with an exact symmetry observation. In the outer strip, a single point $s = \sigma + i\tau$ is tested against the one-point bracket $\mathfrak{B}_{\text{eff}}(\sigma, \tau)$. Near the critical line this is not the correct perspective. A zero away from the critical line never occurs in isolation: by the functional equation and conjugation symmetry it belongs to a reflected pair.

We write

$$s_-(u, \tau) := \frac{1}{2} - u + i\tau, \quad s_+(u, \tau) := \frac{1}{2} + u + i\tau,$$

where

$$0 < u \leq u_0.$$

In the certified inner strip we shall take

$$u_0 = 0.01.$$

Lemma 3.1 (Reflected zero-pair symmetry). *Let*

$$0 < u < \frac{1}{2}, \quad \tau > 0.$$

If

$$s_- = \frac{1}{2} - u + i\tau$$

is a nontrivial zero of $\zeta(s)$, then

$$s_+ = \frac{1}{2} + u + i\tau$$

is also a nontrivial zero of $\zeta(s)$.

Proof. The nontrivial zeros of $\zeta(s)$ are invariant under conjugation

$$s \mapsto \bar{s}$$

and under the functional-equation reflection

$$s \mapsto 1 - s.$$

Equivalently, they are invariant under the combined reflection

$$s \mapsto 1 - \bar{s}.$$

Applying this map to

$$s_- = \frac{1}{2} - u + i\tau$$

gives

$$1 - \bar{s}_- = 1 - \left(\frac{1}{2} - u - i\tau \right) = \frac{1}{2} + u + i\tau = s_+.$$

Therefore s_+ is also a nontrivial zero. □

The quotient numerator was normalized in the structural paper so that every nontrivial zero satisfies

$$\mathcal{N}_Q(s) = 0.$$

The preceding symmetry therefore gives two exact equations from one off-critical zero.

Corollary 3.2 (Two quotient equations at a reflected pair). *If*

$$s_- = \frac{1}{2} - u + i\tau$$

is a nontrivial zero with $0 < u < 1/2$, then

$$\mathcal{N}_Q(s_-) = 0, \quad \mathcal{N}_Q(s_+) = 0,$$

where

$$s_+ = \frac{1}{2} + u + i\tau.$$

Proof. By Lemma 3.1, s_+ is also a nontrivial zero. The quotient numerator satisfies $\mathcal{N}_Q(s) = 0$ at every nontrivial zero. Applying this to both s_- and s_+ proves the claim. $\square$

Remark 3.3 (Why this matters). *The outer-strip proof uses the single equation*

$$\mathcal{N}_Q(s) = 0$$

at one point. In the inner strip, an off-critical zero gives the stronger pair of equations

$$\mathcal{N}_Q(s_-) = 0, \quad \mathcal{N}_Q(s_+) = 0.$$

The two-channel detector below is simply the average and the finite-difference form of these two equations.

4 The two-channel detector

We now define the reflected detector used throughout the inner-strip argument. It is useful to think of the two components as follows:

- the even channel measures the reflected midpoint value of the quotient numerator;
- the divided odd channel measures the reflected transverse slope of the quotient numerator.

At an off-critical reflected zero pair, both quantities must vanish.

Definition 4.1 (Even and divided odd quotient channels). *Let*

$$s_- = \frac{1}{2} - u + i\tau, \quad s_+ = \frac{1}{2} + u + i\tau,$$

with

$$0 < u \leq u_0.$$

Define the even quotient channel by

$$\mathcal{E}(u, \tau) := \frac{\mathcal{N}_Q(s_-) + \mathcal{N}_Q(s_+)}{2},$$

and the divided odd quotient channel by

$$\mathcal{O}(u, \tau) := \frac{\mathcal{N}_Q(s_-) - \mathcal{N}_Q(s_+)}{2u}.$$

The reflected two-channel detector is

$$\mathfrak{D}(u, \tau) := (\mathcal{E}(u, \tau), \mathcal{O}(u, \tau)).$$

Lemma 4.2 (Detector vanishing at an off-critical reflected zero pair). *If $s_- = \frac{1}{2} - u + i\tau$ is a nontrivial zero with $0 < u < 1/2$, then*

$$\mathfrak{D}(u, \tau) = (0, 0).$$

Proof. By [Corollary 3.2](#),

$$\mathcal{N}_Q(s_-) = 0, \quad \mathcal{N}_Q(s_+) = 0.$$

Therefore

$$\mathcal{E}(u, \tau) = \frac{0 + 0}{2} = 0,$$

and

$$\mathcal{O}(u, \tau) = \frac{0 - 0}{2u} = 0.$$

Thus

$$\mathfrak{D}(u, \tau) = (0, 0).$$

□

4.1 Bracket and residual channels

We now insert the structural expansion from [Assumption 2.1](#) into the two reflected channels. For

$$s_{\pm} = \frac{1}{2} \pm u + i\tau,$$

the expansion gives

$$\mathcal{N}_Q(s_{\pm}) = \frac{i}{\tau} \mathfrak{B}_{\text{eff}}\left(\frac{1}{2} \pm u, \tau\right) + \mathcal{R}_{\text{total}}(s_{\pm}).$$

Define the even bracket channel by

$$B_{\text{ev}}(u, \tau) := \frac{\mathfrak{B}_{\text{eff}}(\frac{1}{2} - u, \tau) + \mathfrak{B}_{\text{eff}}(\frac{1}{2} + u, \tau)}{2},$$

and the divided odd bracket channel by

$$B_{\text{od}}(u, \tau) := \frac{\mathfrak{B}_{\text{eff}}(\frac{1}{2} - u, \tau) - \mathfrak{B}_{\text{eff}}(\frac{1}{2} + u, \tau)}{2u}.$$

The bracket vector is

$$\mathfrak{B}(u, \tau) := (B_{\text{ev}}(u, \tau), B_{\text{od}}(u, \tau)).$$

Similarly, define the even residual channel by

$$R_{\text{ev}}(u, \tau) := \frac{\mathcal{R}_{\text{total}}(s_-) + \mathcal{R}_{\text{total}}(s_+)}{2},$$

and the divided odd residual channel by

$$R_{\text{od}}(u, \tau) := \frac{\mathcal{R}_{\text{total}}(s_-) - \mathcal{R}_{\text{total}}(s_+)}{2u}.$$

The residual vector is

$$\mathfrak{R}(u, \tau) := (R_{\text{ev}}(u, \tau), R_{\text{od}}(u, \tau)).$$

Proposition 4.3 (Two-channel bracket-residual decomposition). *Assume [Assumption 2.1](#). For*

$$0 < u \leq u_0, \quad \tau \geq T_0,$$

one has

$$\mathfrak{D}(u, \tau) = \frac{i}{\tau} \mathfrak{B}(u, \tau) + \mathfrak{R}(u, \tau).$$

Equivalently,

$$\mathcal{E}(u, \tau) = \frac{i}{\tau} B_{\text{ev}}(u, \tau) + R_{\text{ev}}(u, \tau),$$

and

$$\mathcal{O}(u, \tau) = \frac{i}{\tau} B_{\text{od}}(u, \tau) + R_{\text{od}}(u, \tau).$$

Proof. Using [Assumption 2.1](#) at s_- and s_+ , we obtain

$$\mathcal{N}_Q(s_-) = \frac{i}{\tau} \mathfrak{B}_{\text{eff}}\left(\frac{1}{2} - u, \tau\right) + \mathcal{R}_{\text{total}}(s_-),$$

and

$$\mathcal{N}_Q(s_+) = \frac{i}{\tau} \mathfrak{B}_{\text{eff}}\left(\frac{1}{2} + u, \tau\right) + \mathcal{R}_{\text{total}}(s_+).$$

Adding these two identities and dividing by 2 gives

$$\mathcal{E}(u, \tau) = \frac{i}{\tau} \frac{\mathfrak{B}_{\text{eff}}(\frac{1}{2} - u, \tau) + \mathfrak{B}_{\text{eff}}(\frac{1}{2} + u, \tau)}{2} + \frac{\mathcal{R}_{\text{total}}(s_-) + \mathcal{R}_{\text{total}}(s_+)}{2}.$$

By definition, this is

$$\mathcal{E}(u, \tau) = \frac{i}{\tau} B_{\text{ev}}(u, \tau) + R_{\text{ev}}(u, \tau).$$

Subtracting the two quotient expansions and dividing by $2u$ gives

$$\mathcal{O}(u, \tau) = \frac{i}{\tau} \frac{\mathfrak{B}_{\text{eff}}(\frac{1}{2} - u, \tau) - \mathfrak{B}_{\text{eff}}(\frac{1}{2} + u, \tau)}{2u} + \frac{\mathcal{R}_{\text{total}}(s_-) - \mathcal{R}_{\text{total}}(s_+)}{2u}.$$

By definition, this is

$$\mathcal{O}(u, \tau) = \frac{i}{\tau} B_{\text{od}}(u, \tau) + R_{\text{od}}(u, \tau).$$

Combining the two component identities proves the vector identity. $\square$

Remark 4.4 (The divided odd channel is harmless only after derivative certification). *The expression*

$$\frac{\mathcal{R}_{\text{total}}(s_-) - \mathcal{R}_{\text{total}}(s_+)}{2u}$$

is not controlled uniformly by an ordinary residual bound alone. Uniformity as $u \rightarrow 0$ requires a bound for

$$\partial_\sigma \mathcal{R}_{\text{total}}(\sigma + i\tau).$$

This is why the inner-strip certificate must include σ -derivative constants for every residual sector.

4.2 The contradiction mechanism

The proof of the inner-strip exclusion will have the following form. Suppose that an off-critical reflected zero pair exists. By [Lemma 4.2](#),

$$\mathfrak{D}(u, \tau) = (0, 0).$$

On the other hand, by [Proposition 4.3](#),

$$\mathfrak{D}(u, \tau) = \frac{i}{\tau} \mathfrak{B}(u, \tau) + \mathfrak{R}(u, \tau).$$

Therefore

$$\|\mathfrak{D}(u, \tau)\| \geq \frac{\|\mathfrak{B}(u, \tau)\|}{\tau} - \|\mathfrak{R}(u, \tau)\|.$$

If we prove

$$\|\mathfrak{B}(u, \tau)\| \geq g_0$$

and

$$\|\mathfrak{R}(u, \tau)\| \leq C_{\text{inner}} \frac{\log \tau}{\tau^{3/2}},$$

then

$$\|\mathfrak{D}(u, \tau)\| \geq \frac{g_0}{\tau} - C_{\text{inner}} \frac{\log \tau}{\tau^{3/2}}.$$

The right-hand side is positive whenever

$$\sqrt{\tau} > \frac{C_{\text{inner}}}{g_0} \log \tau.$$

This contradicts

$$\mathfrak{D}(u, \tau) = (0, 0).$$

Thus the entire inner-strip problem reduces to two tasks:

- (i) prove or certify the bracket-vector lower bound

$$\|\mathfrak{B}(u, \tau)\| \geq g_0;$$

- (ii) prove or certify the residual-vector upper bound

$$\|\mathfrak{R}(u, \tau)\| \leq C_{\text{inner}} \frac{\log \tau}{\tau^{3/2}}.$$

The next sections carry out these two reductions.

5 Bracket-vector lower bound

The purpose of this section is to prove that the reflected bracket vector does not degenerate at the critical line. This is the main structural reason the inner-strip argument is different from the outer-strip argument.

The one-channel bracket

$$\mathfrak{B}_{\text{eff}}(\sigma, \tau)$$

has a lower bound away from $\sigma = 1/2$, but that lower bound degenerates as $\sigma \rightarrow 1/2$. The two-channel bracket vector keeps both the value and the transverse derivative at the critical line. These two quantities cannot vanish simultaneously.

Recall the effective bracket

$$\mathfrak{B}_{\text{eff}}(\sigma, \tau) = 2^\sigma e^{iL} \left(\frac{12 - \sigma}{48} + \frac{iX}{4} \right) - 2^{-\sigma} e^{-iL} \left(\frac{11 + \sigma}{24} - \frac{iX}{2} \right),$$

where

$$L = \tau \log 2, \quad X = \Xi_{\text{end}}(\tau) \in \mathbb{R}.$$

In this section L is treated as an arbitrary real phase and X as an arbitrary real parameter.

5.1 Critical-line bracket value and derivative

Define the critical-line bracket value

$$B_0(L, X) := \mathfrak{B}_{\text{eff}}\left(\frac{1}{2}, \tau\right),$$

and the critical-line transverse derivative

$$B_1(L, X) := \partial_\sigma \mathfrak{B}_{\text{eff}}(\sigma, \tau) \Big|_{\sigma=1/2}.$$

Lemma 5.1 (Critical-line bracket value). *For every real L and every real X ,*

$$B_0(L, X) = \frac{i\sqrt{2}}{48} (24X \cos L + 23 \sin L).$$

In particular, $B_0(L, X)$ is purely imaginary.

Proof. Substitute $\sigma = 1/2$ into the bracket formula. Since

$$2^{1/2} = \sqrt{2}, \quad 2^{-1/2} = \frac{1}{\sqrt{2}},$$

we get

$$B_0(L, X) = \sqrt{2} e^{iL} \left(\frac{23}{96} + \frac{iX}{4} \right) - \frac{1}{\sqrt{2}} e^{-iL} \left(\frac{23}{48} - \frac{iX}{2} \right).$$

Putting both terms over the same denominator gives

$$B_0(L, X) = \frac{\sqrt{2}}{96} [e^{iL}(23 + 24iX) - e^{-iL}(46 - 48iX)].$$

Expanding into sine and cosine terms and simplifying gives

$$B_0(L, X) = \frac{i\sqrt{2}}{48} (24X \cos L + 23 \sin L).$$

□

Lemma 5.2 (Critical-line bracket derivative). *For every real L and every real X ,*

$$B_1(L, X) = \frac{\sqrt{2}}{48} [(23 \log 2 - 2) \cos L - 24X \log 2 \sin L].$$

In particular, $B_1(L, X)$ is real.

Proof. Differentiate the bracket formula with respect to σ . The first bracket component gives

$$\partial_\sigma \left[2^\sigma e^{iL} \left(\frac{12 - \sigma}{48} + \frac{iX}{4} \right) \right] = 2^\sigma e^{iL} \left[\log 2 \left(\frac{12 - \sigma}{48} + \frac{iX}{4} \right) - \frac{1}{48} \right].$$

The second bracket component gives

$$\partial_\sigma \left[-2^{-\sigma} e^{-iL} \left(\frac{11 + \sigma}{24} - \frac{iX}{2} \right) \right] = 2^{-\sigma} e^{-iL} \left[\log 2 \left(\frac{11 + \sigma}{24} - \frac{iX}{2} \right) - \frac{1}{24} \right].$$

Substituting $\sigma = 1/2$, collecting the e^{iL} and e^{-iL} terms, and simplifying yields

$$B_1(L, X) = \frac{\sqrt{2}}{48} [(23 \log 2 - 2) \cos L - 24X \log 2 \sin L].$$

The displayed expression is real because L , X , and $\log 2$ are real. $\square$

5.2 Critical-line vector nondegeneracy

We now prove that B_0 and B_1 cannot vanish simultaneously. More precisely, their combined norm has a uniform positive lower bound.

Let

$$\ell := \log 2, \quad a := 23\ell - 2.$$

Since

$$0.69 < \ell < 0.70,$$

we have

$$a > 0.$$

Proposition 5.3 (Critical-line bracket-vector lower bound). *For every real L and every real X ,*

$$|B_0(L, X)|^2 + |B_1(L, X)|^2 \geq \left(\frac{\sqrt{2}}{48} (23 \log 2 - 2) \right)^2.$$

In particular,

$$|B_0(L, X)|^2 + |B_1(L, X)|^2 > 0.$$

Proof. By [Lemmas 5.1](#) and [5.2](#),

$$|B_0(L, X)|^2 + |B_1(L, X)|^2 = \frac{2}{48^2} [A(L, X)^2 + D(L, X)^2],$$

where

$$A(L, X) := 24X \cos L + 23 \sin L,$$

and

$$D(L, X) := (23\ell - 2) \cos L - 24X\ell \sin L.$$

Set

$$c := \cos L, \quad s := \sin L, \quad a := 23\ell - 2.$$

Then

$$A = 24Xc + 23s, \quad D = ac - 24X\ell s.$$

For fixed L , the quantity

$$A^2 + D^2$$

is a nonnegative quadratic polynomial in X . Its minimum over real X is obtained by elementary projection. A direct calculation gives

$$\min_{X \in \mathbb{R}} (A^2 + D^2) = \frac{(23\ell - 2\cos^2 L)^2}{\ell^2 + (1 - \ell^2)\cos^2 L}.$$

Writing

$$t := \cos^2 L, \quad 0 \leq t \leq 1,$$

this minimum becomes

$$\frac{(23\ell - 2t)^2}{\ell^2 + (1 - \ell^2)t}.$$

We claim that

$$\frac{(23\ell - 2t)^2}{\ell^2 + (1 - \ell^2)t} \geq (23\ell - 2)^2 = a^2$$

for $0 \leq t \leq 1$.

Indeed, since the denominator

$$\ell^2 + (1 - \ell^2)t$$

is positive, it is enough to show

$$(23\ell - 2t)^2 - (23\ell - 2)^2 [\ell^2 + (1 - \ell^2)t] \geq 0.$$

The left-hand side factors as

$$(t - 1)(529\ell^4 - 92\ell^3 - 525\ell^2 + 4t).$$

Since $0 \leq t \leq 1$, we have

$$t - 1 \leq 0.$$

Also, since $0 \leq t \leq 1$, $0.69 < \ell < 0.70$, and all coefficients are explicit, we have the rigorous upper bound

$$\begin{aligned} 529\ell^4 - 92\ell^3 - 525\ell^2 + 4t &\leq 529(0.70)^4 - 92(0.69)^3 - 525(0.69)^2 + 4 \\ &< 0. \end{aligned}$$

Hence

$$t - 1 \leq 0$$

and the second factor is strictly negative. Their product is therefore nonnegative. Therefore

$$A^2 + D^2 \geq a^2.$$

Consequently,

$$|B_0(L, X)|^2 + |B_1(L, X)|^2 \geq \frac{2}{48^2} a^2 = \left(\frac{\sqrt{2}}{48} (23 \log 2 - 2) \right)^2.$$

This proves the proposition. □

Remark 5.4 (Interpretation). *The outer-strip one-channel lower bound degenerates because*

$$|\mathfrak{B}_{\text{eff}}(\sigma, \tau)|$$

can become small as $\sigma \rightarrow 1/2$. The proposition says that the pair

$$\left(\mathfrak{B}_{\text{eff}} \left(\frac{1}{2}, \tau \right), \partial_\sigma \mathfrak{B}_{\text{eff}} \left(\frac{1}{2}, \tau \right) \right)$$

does not degenerate. Thus a reflected zero pair would have to cancel not only the bracket value at the critical line but also its transverse slope. The bracket algebra forbids this simultaneous cancellation.

5.3 Finite-width bracket certificate

The previous proposition proves the limiting lower bound at $u = 0$. The inner-strip detector uses the finite-width bracket vector

$$\mathfrak{B}(u, \tau) = (B_{\text{ev}}(u, \tau), B_{\text{od}}(u, \tau)),$$

where

$$0 < u \leq 0.01.$$

As $u \rightarrow 0$,

$$B_{\text{ev}}(u, \tau) \rightarrow B_0(L, X),$$

and

$$B_{\text{od}}(u, \tau) \rightarrow -B_1(L, X).$$

The critical-line lower bound is approximately

$$\frac{\sqrt{2}}{48}(23 \log 2 - 2) = 0.4107814619 \dots$$

For the public theorem, we use the deliberately smaller finite-width target

$$g_0 = 0.2.$$

The finite-width inequality

$$\|\mathfrak{B}(u, \tau)\| \geq 0.2$$

for

$$0 < u \leq 0.01$$

is certified by a finite real interval calculation as follows. The phase L is reduced modulo 2π . For fixed u and L , the squared norm

$$\|\mathfrak{B}(u, \tau)\|^2$$

is an explicit quadratic polynomial in the real parameter

$$X = \Xi_{\text{end}}(\tau).$$

Thus the dependence on $X \in \mathbb{R}$ is removed by minimizing this quadratic analytically over X . The remaining verification is a compact two-variable interval problem in

$$0 \leq u \leq 0.01, \quad 0 \leq L \leq 2\pi.$$

The public certificate proves the conservative bound

$$\|\mathfrak{B}(u, \tau)\| \geq 0.2.$$

Assumption 5.5 (Finite inner bracket-vector certificate). *The public file*

`inner_bracket_vector_certificate.json`

has status

proved

and certifies

$$\|\mathfrak{B}(u, \tau)\| \geq g_0$$

for

$$0 < u \leq 0.01, \quad \tau \geq T_0,$$

with

$$g_0 = 0.2.$$

Remark 5.6 (Why a finite certificate is acceptable here). *The limiting lower bound at $u = 0$ is analytic. Passing from $u = 0$ to $0 < u \leq 0.01$ is a finite compact verification problem for an explicit elementary expression. No zeta-function estimates, no stationary phase, and no residual analysis enter this bracket-vector certificate. It is a small real interval problem, separate from the much larger residual certificate.*

Proposition 5.7 (Certified inner bracket-vector lower bound). *Assume [Assumption 5.5](#). Then, for every*

$$0 < u \leq 0.01, \quad \tau \geq T_0,$$

one has

$$\|\mathfrak{B}(u, \tau)\| \geq 0.2.$$

Proof. This is exactly the statement certified by [Assumption 5.5](#). □

6 Residual-vector bound

We now turn from the bracket vector to the residual vector. This is the place where the inner-strip argument genuinely differs from the outer-strip argument.

In the outer strip, it is enough to certify the residual itself:

$$|\mathcal{R}_{\text{total}}(s)| \leq C \frac{\log \tau}{\tau^{3/2}}.$$

For the inner reflected detector, this controls the even channel but not the divided odd channel. The odd channel contains the quotient

$$\frac{\mathcal{R}_{\text{total}}(s_-) - \mathcal{R}_{\text{total}}(s_+)}{2u}.$$

A uniform bound for this expression as $u \rightarrow 0$ requires a bound for

$$\partial_\sigma \mathcal{R}_{\text{total}}(\sigma + i\tau).$$

Thus the inner-strip certificate must include both ordinary residual constants and σ -derivative residual constants.

6.1 Sector-level assumptions

For clarity we restate the sector certificate interface in the form used in this section. The total residual is decomposed as

$$\mathcal{R}_{\text{total}} = \sum_{\nu \in \mathcal{S}} \mathcal{R}_\nu,$$

where

$$\mathcal{S} := \{\text{band, bd, far, end, nonstat, floor, core}\}.$$

Assumption 6.1 (Inner residual value and derivative certificates). *For each*

$$\nu \in \mathcal{S},$$

there exist constants

$$C_{\nu,0} \geq 0, \quad C_{\nu,1} \geq 0,$$

such that, for every

$$\left| \sigma - \frac{1}{2} \right| \leq 0.01, \quad \tau \geq T_0,$$

one has

$$|\mathcal{R}_\nu(\sigma + i\tau)| \leq C_{\nu,0} \frac{\log \tau}{\tau^{3/2}},$$

and

$$|\partial_\sigma \mathcal{R}_\nu(\sigma + i\tau)| \leq C_{\nu,1} \frac{\log \tau}{\tau^{3/2}}.$$

Define

$$C_0 := \sum_{\nu \in \mathcal{S}} C_{\nu,0},$$

and

$$C_1 := \sum_{\nu \in \mathcal{S}} C_{\nu,1}.$$

Finally define

$$C_{\text{inner}} := \sqrt{C_0^2 + C_1^2}.$$

Remark 6.2 (Why the constants are summed before taking the vector norm). *The residual vector has two components. The even component is bounded by the sum of the ordinary sector constants, and the divided odd component is bounded by the sum of the derivative sector constants. Only after the two scalar bounds are obtained do we take the Euclidean vector norm. This is why*

$$C_{\text{inner}} = \sqrt{C_0^2 + C_1^2}$$

is the correct inner residual constant.

6.2 Even residual channel

Recall

$$R_{\text{ev}}(u, \tau) = \frac{\mathcal{R}_{\text{total}}(s_-) + \mathcal{R}_{\text{total}}(s_+)}{2},$$

where

$$s_- = \frac{1}{2} - u + i\tau, \quad s_+ = \frac{1}{2} + u + i\tau.$$

Lemma 6.3 (Even residual bound). *Assume [Assumption 6.1](#). For*

$$0 < u \leq 0.01, \quad \tau \geq T_0,$$

one has

$$|R_{\text{ev}}(u, \tau)| \leq C_0 \frac{\log \tau}{\tau^{3/2}}.$$

Proof. By the triangle inequality,

$$|R_{\text{ev}}(u, \tau)| \leq \frac{1}{2} |\mathcal{R}_{\text{total}}(s_-)| + \frac{1}{2} |\mathcal{R}_{\text{total}}(s_+)|.$$

Since

$$\mathcal{R}_{\text{total}} = \sum_{\nu \in \mathcal{S}} \mathcal{R}_\nu,$$

we have

$$|\mathcal{R}_{\text{total}}(s_\pm)| \leq \sum_{\nu \in \mathcal{S}} |\mathcal{R}_\nu(s_\pm)|.$$

Both s_- and s_+ lie in the certified inner strip

$$\left| \sigma - \frac{1}{2} \right| \leq 0.01.$$

Therefore, by [Assumption 6.1](#),

$$|\mathcal{R}_\nu(s_\pm)| \leq C_{\nu,0} \frac{\log \tau}{\tau^{3/2}}.$$

Summing over sectors gives

$$|\mathcal{R}_{\text{total}}(s_\pm)| \leq C_0 \frac{\log \tau}{\tau^{3/2}}.$$

Hence

$$|R_{\text{ev}}(u, \tau)| \leq C_0 \frac{\log \tau}{\tau^{3/2}}.$$

□

6.3 Divided odd residual channel

The divided odd residual channel is

$$R_{\text{od}}(u, \tau) = \frac{\mathcal{R}_{\text{total}}(s_-) - \mathcal{R}_{\text{total}}(s_+)}{2u}.$$

The point of certifying $\partial_\sigma \mathcal{R}_{\text{total}}$ is exactly to control this quantity uniformly as $u \rightarrow 0$.

Lemma 6.4 (Divided odd residual bound). *Assume [Assumption 6.1](#). For*

$$0 < u \leq 0.01, \quad \tau \geq T_0,$$

one has

$$|R_{\text{od}}(u, \tau)| \leq C_1 \frac{\log \tau}{\tau^{3/2}}.$$

Proof. For fixed τ , define

$$F(\sigma) := \mathcal{R}_{\text{total}}(\sigma + i\tau).$$

Then

$$R_{\text{od}}(u, \tau) = \frac{F(\frac{1}{2} - u) - F(\frac{1}{2} + u)}{2u}.$$

By the mean value theorem applied to the real variable σ ,

$$|R_{\text{od}}(u, \tau)| \leq \sup_{|\sigma - \frac{1}{2}| \leq u} |\partial_\sigma \mathcal{R}_{\text{total}}(\sigma + i\tau)|.$$

Since $u \leq 0.01$, the supremum is taken inside the certified inner strip

$$\left| \sigma - \frac{1}{2} \right| \leq 0.01.$$

Using

$$\mathcal{R}_{\text{total}} = \sum_{\nu \in \mathcal{S}} \mathcal{R}_\nu,$$

we have

$$|\partial_\sigma \mathcal{R}_{\text{total}}(\sigma + i\tau)| \leq \sum_{\nu \in \mathcal{S}} |\partial_\sigma \mathcal{R}_\nu(\sigma + i\tau)|.$$

By [Assumption 6.1](#),

$$|\partial_\sigma \mathcal{R}_\nu(\sigma + i\tau)| \leq C_{\nu,1} \frac{\log \tau}{\tau^{3/2}}.$$

Therefore

$$|\partial_\sigma \mathcal{R}_{\text{total}}(\sigma + i\tau)| \leq C_1 \frac{\log \tau}{\tau^{3/2}}.$$

This proves

$$|R_{\text{od}}(u, \tau)| \leq C_1 \frac{\log \tau}{\tau^{3/2}}.$$

□

Remark 6.5 (No loss as $u \rightarrow 0$). *The divided odd residual channel contains $1/u$, but the mean value theorem removes this apparent singularity. The price is exactly the σ -derivative residual certificate. This is why the inner-strip certificate is finite but strictly stronger than the outer-strip certificate.*

6.4 Residual-vector bound

We now combine the two scalar residual estimates.

Proposition 6.6 (Two-channel residual-vector bound). *Assume [Assumption 6.1](#). For*

$$0 < u \leq 0.01, \quad \tau \geq T_0,$$

the residual vector

$$\mathfrak{R}(u, \tau) = (R_{\text{ev}}(u, \tau), R_{\text{od}}(u, \tau))$$

satisfies

$$\|\mathfrak{R}(u, \tau)\| \leq C_{\text{inner}} \frac{\log \tau}{\tau^{3/2}},$$

where

$$C_{\text{inner}} = \sqrt{C_0^2 + C_1^2}.$$

Proof. By [Lemma 6.3](#),

$$|R_{\text{ev}}(u, \tau)| \leq C_0 \frac{\log \tau}{\tau^{3/2}}.$$

By [Lemma 6.4](#),

$$|R_{\text{od}}(u, \tau)| \leq C_1 \frac{\log \tau}{\tau^{3/2}}.$$

Therefore, using the Euclidean norm on $\mathbb{C}^2$,

$$\begin{aligned} \|\mathfrak{R}(u, \tau)\| &= (|R_{\text{ev}}(u, \tau)|^2 + |R_{\text{od}}(u, \tau)|^2)^{1/2} \\ &\leq (C_0^2 + C_1^2)^{1/2} \frac{\log \tau}{\tau^{3/2}} \\ &= C_{\text{inner}} \frac{\log \tau}{\tau^{3/2}}. \end{aligned}$$

□

6.5 Certificate form of the residual-vector theorem

For publication and reproducibility, the constants in [Proposition 6.6](#) are not to be copied by hand from separate calculations. They must be produced by a strict merge of the sector certificates.

Assumption 6.7 (Final inner residual certificate). *The file*

INNER_STRIP_CERTIFICATE_SAFE.json

contains, for every

$$\nu \in \mathcal{S},$$

a dependency file with status

proved

and with fields

$$C_value_upper_safe, \quad C_sigma_derivative_upper_safe.$$

The final file has status

inner_full_certificate

and records the merged values

$$C_0, \quad C_1, \quad C_{\text{inner}}.$$

Corollary 6.8 (Certified residual-vector estimate). *Assume [Assumption 6.7](#). Then the residual vector satisfies*

$$\|\Re(u, \tau)\| \leq C_{\text{inner}} \frac{\log \tau}{\tau^{3/2}}$$

for every

$$0 < u \leq 0.01, \quad \tau \geq T_0.$$

Proof. The final certificate supplies the sector constants satisfying [Assumption 6.1](#). The result then follows from [Proposition 6.6](#). $\square$

Remark 6.9 (What the final certificate must refuse). *A final certificate is not acceptable if any sector lacks either its ordinary value constant or its σ -derivative constant. In particular, a file which certifies only*

$$|\mathcal{R}_\nu(s)|$$

but not

$$|\partial_\sigma \mathcal{R}_\nu(s)|$$

does not certify the inner-strip theorem. The divided odd channel requires the derivative data.

7 Finite certifiability of the derivative residual sectors

The previous section showed that the inner-strip residual vector is controlled once every residual sector is certified both in value and in σ -derivative. We now explain why these derivative certificates are finite in exactly the same sense as the outer-strip residual certificate.

The point of this section is not to reproduce every numerical enclosure. Those belong to the public certificate files. The point is to show that the σ -derivative operation does not introduce any new analytic non-compactness, any new stationary phase family, or any untracked sector. It only differentiates explicit amplitudes, explicit algebraic coefficients, and the explicit quotient projection.

Thus the inner-strip derivative certificate is a finite enhancement of the outer-strip certificate, not a new asymptotic theory.

7.1 General derivative principle

Each residual sector in the quotient expansion has the schematic form

$$\mathcal{R}_\nu(s) = \Pi_Q(s) \mathcal{A}_\nu(s)$$

or a finite sum of such terms, where

$$\Pi_Q(s) = \frac{i\tau}{(1-s)(s+1)}$$

is the quotient projection and $\mathcal{A}_\nu(s)$ is a sector amplitude, integral, or grouped mode sum.

The σ -derivative is therefore

$$\partial_\sigma \mathcal{R}_\nu(s) = (\partial_\sigma \Pi_Q(s)) \mathcal{A}_\nu(s) + \Pi_Q(s) \partial_\sigma \mathcal{A}_\nu(s),$$

with

$$\partial_\sigma \Pi_Q(s) = \Pi_Q(s) \left(\frac{1}{1-s} - \frac{1}{s+1} \right).$$

Equivalently,

$$\partial_\sigma \Pi_Q(s) = \frac{2s i\tau}{[(1-s)(s+1)]^2}.$$

Remark 7.1 (Projection derivative is mandatory). *A derivative certificate which differentiates the sector amplitudes but omits*

$$\partial_\sigma \Pi_Q(s)$$

does not certify the inner-strip theorem. The quotient projection depends on σ , and its derivative is part of the residual derivative.

On the inner strip

$$\left| \sigma - \frac{1}{2} \right| \leq 0.01, \quad \tau \geq 10,$$

the projection and its σ -derivative are elementary rational functions of s and are uniformly bounded by explicit powers of $1/\tau$. Thus the projection derivative introduces no new analytic difficulty.

7.2 Algebraic coefficient derivatives

The two main J -side coefficients are

$$s(s-2)$$

and

$$(1-s)(s+1).$$

Their σ -derivatives are

$$\partial_\sigma [s(s-2)] = 2s-2,$$

and

$$\partial_\sigma [(1-s)(s+1)] = -2s.$$

These are explicit algebraic factors. On the inner strip they have size $O(\tau)$, whereas the original quadratic coefficients have size $O(\tau^2)$. Therefore differentiating the coefficients never worsens the asymptotic order of any sector after the same quotient normalization is applied.

Remark 7.2 (No hidden coefficient sector). *The coefficient derivatives are assigned to the same sector as the undifferentiated coefficient. They are not a new residual type and do not create an eighth sector.*

7.3 Compact-boundary sector

The compact-boundary sector is the delicate sector in the outer-strip certificate. It is also the most important sector for the derivative certificate.

In the compact-boundary scaling, the relevant amplitudes are built from factors of the form

$$(2 + wy)^{-\sigma-2}$$

and

$$(2 + wy)^{\sigma-3},$$

together with their frozen endpoint counterparts

$$2^{-\sigma-2}, \quad 2^{\sigma-3}.$$

Their σ -derivatives are explicit:

$$\partial_\sigma(2 + wy)^{-\sigma-2} = -\log(2 + wy)(2 + wy)^{-\sigma-2},$$

$$\partial_\sigma(2 + wy)^{\sigma-3} = \log(2 + wy)(2 + wy)^{\sigma-3},$$

and

$$\partial_\sigma 2^{-\sigma-2} = -\log 2 \, 2^{-\sigma-2},$$

$$\partial_\sigma 2^{\sigma-3} = \log 2 \, 2^{\sigma-3}.$$

In the compact-boundary certificate the cutoff satisfies

$$0 \leq y \leq 2Y, \quad Y = 128.$$

Therefore

$$0 \leq y \leq 256.$$

For

$$0 \leq w \leq \frac{1}{\sqrt{10}},$$

the logarithmic factors

$$\log(2 + wy)$$

are bounded on the compact certificate domain. Hence differentiating in σ only inserts bounded logarithmic amplitude factors inside the same finite collection of Arb boxes.

The compact-boundary derivative verifier must therefore enclose, on every box, both the original compact-boundary integrand and its σ -derivative. For a grouped contribution of the form

$$G_\nu(s) = c_+(s)\Delta_+(s) + c_-(s)\Delta_-(s),$$

the derivative is

$$\partial_\sigma G_\nu(s) = c'_+(s)\Delta_+(s) + c_+(s)\partial_\sigma \Delta_+(s) + c'_-(s)\Delta_-(s) + c_-(s)\partial_\sigma \Delta_-(s).$$

If this grouped contribution is then multiplied by the quotient projection, one must also apply

$$\partial_\sigma(\Pi_Q G_\nu) = (\partial_\sigma \Pi_Q)G_\nu + \Pi_Q \partial_\sigma G_\nu.$$

Proposition 7.3 (Finite certifiability of the compact-boundary derivative). *The compact-boundary sector admits a finite σ -derivative certificate in the inner-strip norm*

$$|\partial_\sigma \mathcal{R}_{\text{bd}}(s)| \leq C_{\text{bd},1} \frac{\log \tau}{\tau^{3/2}}.$$

Proof. The compact-boundary sector is already reduced to finitely many compact boxes in the variables

$$\sigma, \quad w, \quad r, \quad y, \quad \lambda.$$

Differentiating with respect to σ leaves the phases unchanged and only differentiates explicit amplitudes, explicit algebraic coefficients, and the quotient projection. The amplitude derivatives insert the bounded factors

$$\log(2 + wy) \quad \text{and} \quad \log 2$$

on the compact support $0 \leq y \leq 256$. The coefficient and projection derivatives are explicit rational functions of s . Therefore the same finite box subdivision certifies the derivative integrands by Arb ball arithmetic. Summing the resulting box enclosures gives a finite constant $C_{\text{bd},1}$. $\square$

7.4 Compact-complement sectors

The compact complement is split into the same three hooks as in the outer-strip certificate:

$$C_{\chi^c} = C_{\chi^c, \text{tail}} + C_{\chi^c, \text{nonstat}} + C_{\chi^c, \text{stationary}}.$$

For the inner strip, each hook receives a derivative companion.

Tail hook

The compact-complement tail hook is controlled by derivative envelopes followed by integrations by parts. The derivative certificate requires envelopes not only for the original amplitude derivatives but also for their σ -derivatives. Schematically, if the original tail hook certifies

$$|H'(y)| \leq M_1, \quad |H''(y)| \leq M_2,$$

then the derivative hook certifies

$$|\partial_\sigma H'(y)| \leq M_{1,\sigma}, \quad |\partial_\sigma H''(y)| \leq M_{2,\sigma}.$$

The same integrations by parts then produce the derivative tail constant

$$C_{\chi^c, \text{tail}, 1}.$$

Nonstationary hook

The nonstationary hook is based on a lower bound for the phase derivative. The phase derivative is independent of σ ; only amplitudes and coefficients are differentiated. Hence the same nonstationary box partition and the same phase-separation condition apply to the derivative integrands. The integration-by-parts bound is repeated with the σ -differentiated amplitudes.

Stationary-overlap hook

The stationary-overlap hook is compact and is certified by direct Arb enclosure. Its derivative version is obtained by enclosing the σ -differentiated integrand on the same compact boxes.

Proposition 7.4 (Finite certifiability of compact-complement derivatives). *Each compact-complement hook admits a finite σ -derivative certificate:*

$$C_{\chi^c, \text{tail}, 1}, \quad C_{\chi^c, \text{nonstat}, 1}, \quad C_{\chi^c, \text{stationary}, 1}.$$

Consequently the full compact-complement derivative contribution is finitely certifiable.

Proof. The tail hook uses derivative-envelope bounds; differentiating with respect to σ replaces the original envelopes by envelopes for the σ -differentiated amplitudes. The nonstationary hook uses the same phase-separation boxes because the phases do not depend on σ . The stationary-overlap hook is compact and is directly enclosed after differentiation. Each operation is performed on a finite box subdivision, and all differentiated amplitudes are explicit elementary functions. Therefore all three derivative constants are finite and certifiable. $\square$

7.5 Band sector

The band sector comes from the quotient stationary range

$$\frac{\tau}{4\pi} < m \leq \frac{\tau}{2\pi}.$$

The relevant sums contain powers such as

$$m^{-s}, \quad m^{-(1-s)}.$$

Differentiating in σ gives

$$\partial_\sigma m^{-s} = -(\log m)m^{-s},$$

and

$$\partial_\sigma m^{-(1-s)} = (\log m)m^{-(1-s)}.$$

On the band,

$$m \asymp \tau,$$

so

$$\log m \leq C \log \tau$$

for an explicit constant C . The Poisson endpoint expansion and the nonstationary dual-frequency estimates therefore remain finite after differentiation, with derivative amplitudes containing at most logarithmic factors.

Proposition 7.5 (Finite certifiability of the band derivative). *The band residual admits a finite σ -derivative certificate*

$$|\partial_\sigma \mathcal{R}_{\text{band}}(s)| \leq C_{\text{band},1} \frac{\log \tau}{\tau^{3/2}}$$

on the certified inner strip.

Proof. The differentiated band sums are obtained from the original band sums by inserting factors $\pm \log m$. In the band range $m \asymp \tau$, these factors are bounded by explicit multiples of $\log \tau$. The same Poisson endpoint decomposition applies. The endpoint stationary contributions are differentiated explicitly, and the nonstationary dual-frequency tails are bounded by the same integration-by-parts scheme with differentiated amplitudes. Thus the band derivative is controlled in the same residual norm, with a finite constant $C_{\text{band},1}$. $\square$

7.6 Endpoint sector

The endpoint sector contains the regularized endpoint scalar

$$X = \Xi_{\text{end}}(\tau).$$

This scalar depends on τ , but not on σ . Therefore ∂_σ acts only on the explicit powers and coefficients in the endpoint bracket and endpoint residual.

The relevant differentiated factors are combinations of

$$2^\sigma, \quad 2^{-\sigma}, \quad 12 - \sigma, \quad 11 + \sigma,$$

whose derivatives are elementary:

$$\partial_\sigma 2^\sigma = (\log 2) 2^\sigma,$$

$$\partial_\sigma 2^{-\sigma} = -(\log 2) 2^{-\sigma},$$

$$\partial_\sigma (12 - \sigma) = -1,$$

and

$$\partial_\sigma (11 + \sigma) = 1.$$

Proposition 7.6 (Finite certifiability of the endpoint derivative). *The endpoint residual admits a finite σ -derivative certificate*

$$|\partial_\sigma \mathcal{R}_{\text{end}}(s)| \leq C_{\text{end},1} \frac{\log \tau}{\tau^{3/2}}.$$

Proof. The endpoint residual is an explicit finite combination of the real scalar $\Xi_{\text{end}}(\tau)$, powers $2^{\pm\sigma}$, and algebraic coefficients in σ . Since Ξ_{end} is independent of σ , the σ -derivative is obtained by differentiating only elementary factors. The resulting expression is explicit and is bounded by the same interval method used for the endpoint sector itself. $\square$

7.7 Far stationary sector

The far stationary sector is controlled away from the endpoint-stationary transition. Its amplitudes depend on σ through powers of the stationary point

$$x_{m,*} = \frac{\tau}{2\pi m}$$

and through explicit algebraic coefficients. Differentiating those powers inserts logarithmic factors such as

$$\log x_{m,*}.$$

In the far stationary range, $x_{m,*}$ stays in the same controlled domain as in the original stationary analysis. The differentiated stationary remainder therefore has the same summability structure, with explicit logarithmic weights.

Proposition 7.7 (Finite certifiability of the far stationary derivative). *The far stationary residual admits a finite σ -derivative certificate*

$$|\partial_\sigma \mathcal{R}_{\text{far}}(s)| \leq C_{\text{far},1} \frac{\log \tau}{\tau^{3/2}}.$$

Proof. The far stationary expansion is uniform away from the boundary transition. Differentiating with respect to σ differentiates only amplitudes and coefficients; the stationary phase location and curvature are independent of σ . The differentiated amplitudes contain logarithmic factors $\log x_{m,*}$, which are explicitly bounded on the far stationary boxes. The same summation and interval enclosure used for the undifferentiated far sector therefore gives a finite derivative constant. $\square$

7.8 Always-nonstationary sector

The always-nonstationary sector is controlled by integration by parts in regions where the phase derivative is separated from zero. The phases do not depend on σ . Therefore the nonstationary separation is unchanged under ∂_σ .

Differentiating the sector only differentiates amplitudes, coefficients, and possibly the quotient projection. The differentiated amplitudes contain bounded logarithmic factors on each certified box.

Proposition 7.8 (Finite certifiability of the nonstationary derivative). *The always-nonstationary residual admits a finite σ -derivative certificate*

$$|\partial_\sigma \mathcal{R}_{\text{nonstat}}(s)| \leq C_{\text{nonstat},1} \frac{\log \tau}{\tau^{3/2}}.$$

Proof. The integration-by-parts denominators are phase derivatives and are independent of σ . Hence the same lower bounds for the phase derivatives apply to the differentiated sector. The σ -derivative acts only on elementary amplitudes, coefficients, and the projection. These differentiated terms are explicit and are enclosed on the same finite partition. Thus the nonstationary derivative sector is finitely certifiable. $\square$

7.9 Floor and core sectors

The floor sector is eliminated by the same half-open convention as in the outer-strip proof. If

$$\mathcal{R}_{\text{floor}} = 0,$$

then also

$$\partial_\sigma \mathcal{R}_{\text{floor}} = 0.$$

The core sector consists of algebraic quotient-normalization corrections and coefficient mismatches. These are explicit functions of s , τ , and the quotient projection. Differentiating the core sector gives explicit rational functions and elementary factors, all bounded on the inner strip.

Proposition 7.9 (Finite certifiability of floor and core derivatives). *The floor and core residuals admit finite derivative certificates. In particular, if the floor sector is identically zero, then*

$$C_{\text{floor},1} = 0.$$

The core derivative satisfies

$$|\partial_\sigma \mathcal{R}_{\text{core}}(s)| \leq C_{\text{core},1} \frac{\log \tau}{\tau^{3/2}}$$

for a finite explicit constant $C_{\text{core},1}$.

Proof. The floor statement is immediate if the floor residual is identically zero. The core sector is algebraic in the quotient coefficients, coefficient ratios, and projection factors. Its σ -derivative is obtained by differentiating explicit rational functions of s . On

$$\left| \sigma - \frac{1}{2} \right| \leq 0.01, \quad \tau \geq 10,$$

these rational functions have explicit interval bounds. Therefore the core derivative is finitely certifiable. $\square$

7.10 Summary of derivative certifiability

Combining the preceding propositions, every residual sector admits a σ -derivative certificate in the inner-strip norm. Therefore the only remaining task is computational: produce the seven sector JSON files with both ordinary and derivative constants, and merge them into the final public certificate.

Theorem 7.10 (Finite certifiability of the inner residual vector). *The inner-strip residual-vector bound*

$$\|\mathfrak{R}(u, \tau)\| \leq C_{\text{inner}} \frac{\log \tau}{\tau^{3/2}}$$

is finitely certifiable from the seven sector certificates

$$\{\text{band}, \text{bd}, \text{far}, \text{end}, \text{nonstat}, \text{floor}, \text{core}\},$$

provided each sector certificate contains both a value constant and a σ -derivative constant.

Proof. By [Propositions 7.3 to 7.9](#), each sector derivative is finitely certifiable. The ordinary sector residuals are the same sector residuals as in the outer-strip certificate. Once the ordinary and derivative constants are available, the residual-vector bound is exactly [Proposition 6.6](#). $\square$

Remark 7.11 (What has been reduced to computation). *The analytic work in this section shows that no new infinite analytic problem appears after differentiating in σ . The derivative certificate is a finite interval-arithmetic problem on the same structural decomposition as the outer-strip certificate. The public proof is complete only after the final certificate file records the corresponding constants.*

8 Certified inner-strip exclusion

We now combine the reflected zero-pair identity, the two-channel bracket-residual decomposition, the bracket-vector lower bound, and the residual-vector estimate.

The proof is formally parallel to the outer-strip exclusion proof, but with two important differences:

- (i) the scalar bracket lower bound is replaced by the vector lower bound

$$\|\mathfrak{B}(u, \tau)\| \geq g_0;$$

- (ii) the scalar residual bound is replaced by the residual-vector bound

$$\|\mathfrak{R}(u, \tau)\| \leq C_{\text{inner}} \frac{\log \tau}{\tau^{3/2}}.$$

The advantage of the two-channel formulation is that it remains meaningful as

$$u \rightarrow 0.$$

Theorem 8.1 (Conditional certified inner-strip exclusion). *Assume the structural quotient expansion*

$$\mathcal{N}_Q(s) = \frac{i}{\tau} \mathfrak{B}_{\text{eff}}(\sigma, \tau) + \mathcal{R}_{\text{total}}(s)$$

from [Assumption 2.1](#). Assume also the finite bracket-vector certificate of [Assumption 5.5](#) and the final inner residual certificate of [Assumption 6.7](#).

Let

$$0 < u \leq 0.01, \quad s_{\pm} = \frac{1}{2} \pm u + i\tau.$$

If $T \geq T_0$ satisfies

$$\sqrt{T} > \frac{C_{\text{inner}}}{g_0} \log T,$$

then there is no reflected off-critical zero pair

$$s_- = \frac{1}{2} - u + i\tau, \quad s_+ = \frac{1}{2} + u + i\tau$$

with

$$\tau \geq T.$$

Equivalently, no nontrivial zero exists in the inner strip

$$0 < \left| \Re s - \frac{1}{2} \right| \leq 0.01, \quad \Im s \geq T.$$

Proof. Suppose, for contradiction, that

$$s_- = \frac{1}{2} - u + i\tau$$

is a nontrivial zero with

$$0 < u \leq 0.01, \quad \tau \geq T.$$

By [Lemma 3.1](#), the reflected point

$$s_+ = \frac{1}{2} + u + i\tau$$

is also a nontrivial zero. Hence, by [Lemma 4.2](#),

$$\mathfrak{D}(u, \tau) = (0, 0).$$

On the other hand, by the two-channel decomposition [Proposition 4.3](#),

$$\mathfrak{D}(u, \tau) = \frac{i}{\tau} \mathfrak{B}(u, \tau) + \mathfrak{R}(u, \tau).$$

Therefore

$$\|\mathfrak{D}(u, \tau)\| \geq \frac{\|\mathfrak{B}(u, \tau)\|}{\tau} - \|\mathfrak{R}(u, \tau)\|.$$

By [Proposition 5.7](#),

$$\|\mathfrak{B}(u, \tau)\| \geq g_0.$$

By [Corollary 6.8](#),

$$\|\mathfrak{R}(u, \tau)\| \leq C_{\text{inner}} \frac{\log \tau}{\tau^{3/2}}.$$

Thus

$$\|\mathfrak{D}(u, \tau)\| \geq \frac{g_0}{\tau} - C_{\text{inner}} \frac{\log \tau}{\tau^{3/2}}.$$

The right-hand side is positive whenever

$$\sqrt{\tau} > \frac{C_{\text{inner}}}{g_0} \log \tau.$$

It remains to pass from the height inequality at T to every $\tau \geq T$. Since

$$t \mapsto \frac{\sqrt{t}}{\log t}$$

is increasing for $t > e^2$, the inequality

$$\sqrt{T} > \frac{C_{\text{inner}}}{g_0} \log T$$

implies

$$\sqrt{\tau} > \frac{C_{\text{inner}}}{g_0} \log \tau$$

for every

$$\tau \geq T, \quad T \geq e^2.$$

Therefore

$$\|\mathfrak{D}(u, \tau)\| > 0.$$

This contradicts

$$\mathfrak{D}(u, \tau) = (0, 0).$$

Hence no such reflected off-critical zero pair exists. $\square$

Remark 8.2 (Why this is genuinely an inner-strip theorem). *The theorem does not rely on a one-point lower bound for*

$$|\mathfrak{B}_{\text{eff}}(\sigma, \tau)|.$$

Instead it uses the reflected vector lower bound

$$\|\mathfrak{B}(u, \tau)\| \geq g_0.$$

This is why the theorem remains stable as $u \rightarrow 0$. The divided odd channel is precisely what captures the transverse derivative information lost by the one-channel outer-strip bracket.

9 Height bridge and public numerical target

The certified height condition is

$$\sqrt{T} > \frac{C_{\text{inner}}}{g_0} \log T.$$

The public bracket-vector certificate uses the conservative value

$$g_0 = 0.2.$$

Therefore the sufficient height condition becomes

$$\sqrt{T} > 5C_{\text{inner}} \log T.$$

For the shared height

$$T = 10^{30},$$

this condition is equivalent to

$$C_{\text{inner}} < 0.2 \frac{\sqrt{10^{30}}}{\log(10^{30})}.$$

Since

$$\sqrt{10^{30}} = 10^{15},$$

and

$$\log(10^{30}) = 30 \log 10,$$

it is enough that

$$C_{\text{inner}} \leq 2.895296546022 \times 10^{12}.$$

Proposition 9.1 (Public height criterion for 10^{30}). *Assume the public bracket-vector lower bound*

$$g_0 = 0.2.$$

If the final inner certificate gives

$$C_{\text{inner}} \leq 2.895296546022 \times 10^{12},$$

then

$$T = 10^{30}$$

is an admissible certified inner-strip height.

Proof. Substituting

$$T = 10^{30}$$

and

$$g_0 = 0.2$$

into the height condition gives

$$10^{15} > \frac{C_{\text{inner}}}{0.2} 30 \log 10.$$

This is equivalent to

$$C_{\text{inner}} < 0.2 \frac{10^{15}}{30 \log 10}.$$

The right-hand side is

$$2.895296546022 \dots \times 10^{12}.$$

Thus the displayed upper bound for C_{inner} implies the height inequality. $\square$

Corollary 9.2 (Certified inner strip at the shared height). *Assume the hypotheses of Theorem 8.1. Assume further that the final public certificate records*

$$g_0 = 0.2$$

and

$$C_{\text{inner}} \leq 2.895296546022 \times 10^{12}.$$

Then there are no nontrivial zeros in

$$0 < \left| \Re s - \frac{1}{2} \right| \leq 0.01, \quad \Im s \geq 10^{30}.$$

Proof. This follows immediately from Theorem 8.1 and Proposition 9.1. $\square$

Remark 9.3 (If the final constant is larger). *If the final certificate gives*

$$C_{\text{inner}} > 2.895296546022 \times 10^{12},$$

then the inner-strip theorem is not invalid. One simply cannot use the shared height 10^{30} with the conservative value $g_0 = 0.2$. The certified height must then be recomputed from

$$\sqrt{T} > \frac{C_{\text{inner}}}{0.2} \log T.$$

10 Public certificate interface

The final inner-strip theorem is a certified theorem only when the public JSON certificate exists and has the required status. The purpose of this section is to make the certificate contract explicit.

The final public certificate is

`INNER_STRIP_CERTIFICATE_SAFE.json.`

It must have

`status = "inner_full_certificate".`

It must also record the following fields:

`u0,      g0_bracket_vector_lower,`
`C_even_residual_upper,      C_sigma_residual_upper,`
`C_inner_vector_upper,      height_tested,`

and

`height_condition_holds.`

The final certificate must depend on seven sector certificates:

`band,   bd,   far,   end,   nonstat,   floor,   core.`

Each sector certificate must have status

`proved`

and must contain both

`C_value_upper_safe`

and

`C_sigma_derivative_upper_safe.`

Definition 10.1 (Acceptable inner certificate). *An inner-strip certificate is acceptable if and only if:*

(i) *the final file*

`INNER_STRIP_CERTIFICATE_SAFE.json`

has

`status = "inner_full_certificate";`

(ii) *the bracket-vector lower field satisfies*

$g0_bracket_vector_lower \geq 0.2;$

(iii) *every sector dependency has status*

`proved;`

(iv) *every sector dependency contains both an ordinary residual constant and a σ -derivative residual constant;*

(v) the final file records

height_condition_holds = true.

Remark 10.2 (No theorem without the final status). *A collection of partial certificates does not certify the inner-strip theorem. The theorem is certified only after the strict merger produces the final status*

inner_full_certificate.

This prevents accidental use of a certificate missing one of the derivative sectors.

10.1 Reproduction commands

The expected public workflow is the following. First create the inner certificate directory:

```
mkdir -p certs_inner
```

Run the bracket-vector verifier:

```
python scripts/cert_inner_bracket_vector.py \
> certs_inner/inner_bracket_vector.raw.json
```

Then run the seven derivative-aware sector verifiers. Their final safe outputs must have the following form:

```
certs_inner/band_sector.safe.json
certs_inner/bd_sector.safe.json
certs_inner/far_sector.safe.json
certs_inner/end_sector.safe.json
certs_inner/nonstat_sector.safe.json
certs_inner/floor_sector.safe.json
certs_inner/core_sector.safe.json
```

Finally run the strict merger:

```
python scripts/merge_inner_certificate_safe.py \
--inputs 'certs_inner/*_sector.safe.json' \
--height 1e30 \
--g0 0.2 \
--output INNER_STRIP_CERTIFICATE_SAFE.json
```

The theorem at height 10^{30} is certified only if the final output has

status = "inner_full_certificate".

11 Combined consequence with the certified outer strip

The certified outer-strip theorem gives the zero-free region

$$0.01 \leq \Re s \leq 0.49, \quad \Im s \geq 10^{30},$$

and, by reflection,

$$0.51 \leq \Re s \leq 0.99, \quad \Im s \geq 10^{30}.$$

The inner-strip theorem certified in this paper covers

$$0 < \left| \Re s - \frac{1}{2} \right| \leq 0.01, \quad \Im s \geq 10^{30},$$

provided the final inner certificate satisfies the public height criterion.

Thus, after both the outer-strip and inner-strip certificates are complete, the combined certified region is

$$0.01 \leq \Re s \leq 0.99, \quad \Re s \neq \frac{1}{2}, \quad \Im s \geq 10^{30}.$$

Corollary 11.1 (Combined certified off-critical exclusion above the shared height). *Assume the certified outer-strip theorem and assume the acceptable inner certificate of [Definition 10.1](#) verifies the height*

$$T = 10^{30}.$$

Then the Riemann zeta function has no nontrivial zeros with

$$0.01 \leq \Re s \leq 0.99, \quad \Re s \neq \frac{1}{2}, \quad \Im s \geq 10^{30}.$$

Proof. The outer-strip theorem covers

$$0.01 \leq \Re s \leq 0.49$$

and its reflected strip

$$0.51 \leq \Re s \leq 0.99.$$

The inner-strip theorem covers

$$0 < \left| \Re s - \frac{1}{2} \right| \leq 0.01,$$

which is precisely

$$0.49 \leq \Re s < \frac{1}{2} \quad \text{and} \quad \frac{1}{2} < \Re s \leq 0.51.$$

Together these regions cover

$$0.01 \leq \Re s \leq 0.99, \quad \Re s \neq \frac{1}{2}.$$

□

Remark 11.2 (Extreme strips near 0 and 1). *The statement above does not claim a certified exclusion in*

$$0 < \Re s < 0.01 \quad \text{or} \quad 0.99 < \Re s < 1.$$

Those extreme strips would require either a separate certified outer-strip run with a smaller left endpoint, or an independent zero-free argument covering those regions.

Remark 11.3 (This is not a finite-height verification below 10^{30}). *The combined theorem is a high-height off-critical exclusion. It does not by itself verify zeros below 10^{30} . To turn the combined high-height exclusion into a full verification statement up to the critical line, one would also need an independent finite-height verification up to the same height. No such finite-height verification is claimed in this paper.*

Remark 11.4 (No claim about zeros on the critical line). *The theorem excludes off-critical zeros in the certified high-height region. It does not exclude zeros on the critical line. The critical line is allowed and is not part of the contradiction.*

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